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SkipperCam: A Wide-Field Skipper-CCD Pathfinder for High-Cadence Imaging with sub-electron capabilities

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Abstract: We present the science program and observational strategy for a Skipper-CCD imager pathfinder designed to demonstrate the potential of ultra-low-noise, high-cadence wide-field imaging. The pathfinder will deploy a 20-sensor Skipper-CCD focal plane on a 0.6 m telescope, enabling sub-electron noise performance combined with rapid readout and a wide field of view. While primarily conceived as a technology demonstrator, this effort is optimized to deliver early scientific results through a dedicated time-domain survey.

The core science program is a high-cadence microlensing survey of the Large Magellan Cloud, targeting short-timescale events produced by low-mass compact objects. The pathfinder will also explore additional science opportunities in fast photometric variability, transient detection, and blue-sensitive imaging, leveraging the intrinsic capabilities of Skipper-CCD technology.

Beyond its immediate science return, the pathfinder establishes a critical experimental platform to validate detector performance, survey strategies, and data acquisition architectures for next-generation wide-field instruments. It will provide the first on-sky demonstration of a multiplexed Skipper-CCD array operating in survey mode, bridging laboratory performance and large-scale deployment. The results will directly inform the design of future facilities, including degree-scale imagers aimed at cosmology, time-domain astrophysics, and multi-messenger science.

This program represents a key step toward enabling a new observational regime defined by photon-limited precision, rapid cadence, and wide-field coverage, with broad implications for astrophysics and fundamental physics.

Contents

1 Introduction

1.1 Motivation: A New Regime in Wide-Field Imaging

Wide-field optical surveys have transformed our understanding of the Universe, enabling precision cosmology, time-domain astrophysics, and the discovery of new classes of transient phenomena. Instruments such as DECam [1], and the LSST [2] have demonstrated the power of combining large étendue with deep imaging to map the dynamic sky. However, despite these advances, current instruments remain limited by detector readout noise and cadence constraints, which restrict sensitivity to faint and rapidly varying signals.

A new observational regime can be achieved by combining wide-field imaging with sub-electron readout noise and rapid observation cadence. In this regime, measurements shift from being limited by detector noise to being limited by photon counts. This allows for the detection of extremely low signal levels and short-timescale variability. Accessing this regime is especially important for scientific cases involving faint, fast, or rare phenomena, where both sensitivity and high temporal resolution are crucial.

1.2 Skipper-CCD Technology

Skipper-CCDs [3] represent a breakthrough in low-noise imaging by enabling multiple non-destructive measurements of the same pixel charge. By averaging repeated samples, the readout noise can be reduced to arbitrarily low levels, reaching well below one electron. This capability has already enabled applications in particle physics and dark matter searches [4, 5], where detection of single or few-electron signals is required.

Recent developments in readout electronics and sensor design [6] have extended the applicability of Skipper-CCDs to astronomical imaging [7]. In particular, the combination of multiplexed readout architectures and optimized operating modes allows these detectors to operate in both fast, moderate-noise regimes and slower, ultra-low-noise regimes. This flexibility opens the possibility of deploying Skipper-CCD technology in wide-field survey instruments, where both cadence and sensitivity are critical.

1.3 Limitations of Current Surveys

Existing wide-field surveys have achieved remarkable performance in both depth and cadence, enabling major advances in time-domain astronomy and cosmology [8, 9]. Modern facilities can operate across a wide range of observing modes, and in many cases reach impressive sensitivity even at relatively short exposure times. However, detector architectures impose fundamental constraints on how noise performance scales with readout speed.

In conventional CCD systems [10], each pixel is measured once per exposure, resulting in a readout noise that is set by the bandwidth of the amplifier. Faster readout requires higher bandwidth, which increases noise, leading to an intrinsic trade-off between cadence and sensitivity. This trade-off is well understood and has shaped the design of current survey instruments.

Skipper-CCDs [3], in their traditional mode of operation, achieve sub-electron noise through multiple non-destructive measurements of each pixel. However, this comes at the cost of significantly increased readout time, making them intrinsically slow when operated in ultra-low-noise regimes. As a result, their application to wide-field, high-cadence astronomy has been limited.

The approach explored in this work addresses this limitation by combining Skipper-CCD technology with a massively parallel readout architecture [11]. By deploying a large number of amplifiers operating simultaneously, the system achieves fast readout of the focal plane while maintaining the ability to perform multiple sampling at the pixel level. In this way, the traditional coupling between low noise and slow readout is mitigated at the system level rather than at the single-amplifier level.

This enables a new observational regime in which wide-field imaging can be performed at high cadence with moderate noise, while preserving the capability to reach sub-electron noise through increased sampling when required. For microlensing surveys targeting low-mass lenses, this combination is particularly powerful, as it allows sensitivity to short-timescale events without sacrificing the ability to detect low-amplitude signals. More broadly, this architecture opens a path toward wide-field instruments that simultaneously approach the limits of cadence, sensitivity, and field of view.

1.4 Pathfinder Concept and Objectives

In this work, we present the science program and observational strategy for a Skipper-CCD imager pathfinder designed to explore this new observational regime. The instrument consists of a 20-sensor Skipper-CCD focal plane (27 Mpix) deployed on a 0.6 m-class telescope, providing a wide field of view of $\sim 2\ \mathrm{deg}^2$ combined with rapid readout and low-noise performance.

While the primary goal of the pathfinder is to demonstrate the performance of Skipper-CCD technology in an on-sky, survey-mode environment, the program is explicitly designed to deliver scientific results. The core science objective is a high-cadence microlensing survey of the LMC, targeting short-timescale events produced by low-mass compact objects. Additional targets such as the Galactic Bulge may also be considered. Complementary science includes fast photometric variability, transient detection, and blue-sensitive imaging.

Beyond its immediate science return, the pathfinder serves as a critical step toward the development of next-generation wide-field instruments based on Skipper-CCD technology. By validating detector performance, survey strategies, and data acquisition systems in a realistic observational setting, this effort reduces technical risk and informs the design of future degree-scale imagers for cosmology, time-domain astrophysics, and multi-messenger science.

A key element of this program is its direct connection to future deployment on the WIYN 3.5 m telescope [12]. The SkipperCam pathfinder is explicitly designed as a scalable prototype of a modular focal plane architecture compatible with the One Degree Imager (ODI) [13]. The detector raft reproduces the mechanical and interface footprint of a 2×2 OTA module in ODI, enabling direct integration within the WIYN focal plane with minimal modification. In this way, the pathfinder functions not only as a technology demonstrator, but also as a risk-reduction platform for a future wide-field Skipper-CCD instrument at WIYN, establishing a clear pathway toward a degree-scale imager on a larger-aperture facility.

The pathfinder instrument will be realized by integrating the 20-CCD SkipperCam focal plane into the existing PreCam camera [14] system and deploying it on the Curtis-Schmidt 0.6 m telescope at CTIO. The PreCam cryostat, originally developed for the Dark Energy Survey[8], provides a well-characterized and fully functional platform including vacuum, cryogenic cooling, shutter, and filter wheel systems. The original DECam detectors have been removed and will be replaced by the

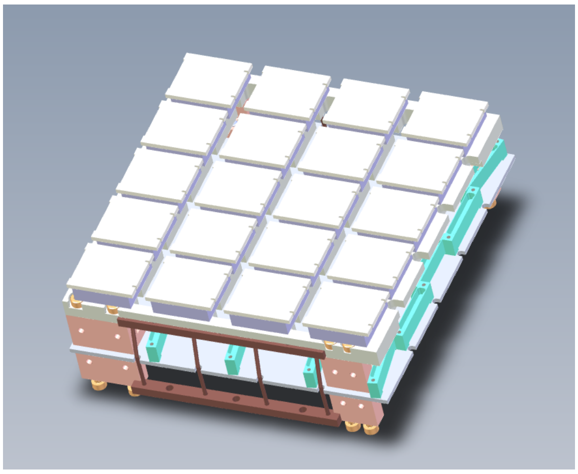

Figure 1. Figure of the focal plane array with 20 CCDs, electronic board, G10 mounting posts and thermal straps

Skipper-CCD raft described above. This approach leverages existing infrastructure to enable rapid deployment while minimizing engineering complexity, allowing the effort to focus on detector performance, readout architecture, and survey operation.

1.5 Paper Overview

The remainder of this paper is organized as follows. Section 2 describes the pathfinder instrument: the focal-plane and detector architecture, the readout and noise performance of the skipper CCDs, the host telescope and optics, and the resulting survey capabilities in field of view, cadence, and depth.

Section 3 presents the science drivers. We first introduce the one-degree imager pathfinder concept (Sec. 3.1), and then develop in detail a fast microlensing survey of the Large Magellanic Cloud (Sec. 3.2), with its detection pipeline, including finite-sources treatment that define the survey's detection efficiency. Using this efficiency we derive constraints on fast-moving and compact lens populations (Sec. 3.3) in a model-independent limit on the microlensing event rate, and mass-dependent limits for a fast-lens benchmark. We also calculate the resulting bounds on primordial black holes as dark matter towards the LMC (Sec. 3.3.3). We then examine how the survey's bright magnitude limit shapes its susceptibility to finite-source suppression and what the fast cadence implies for the characterization of microlensing systems (Sec. 3.4), Finally we briefly discuss the possibility for an LSST-driven stellar occultations program as an additional science case for the instrument (Sec. 3.5). Section 4 outlines the data-management, image-reduction, and photometry strategy needed to operate the survey at its high cadence and data rate. The full instrumental, observing, and target parameters of the survey are laid out in Appendix A.

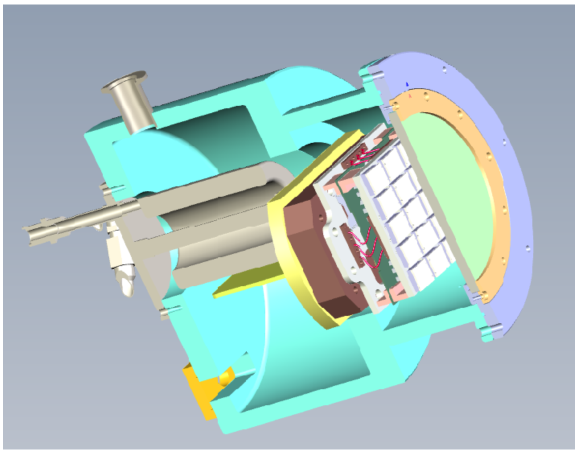

Figure 2. Raft installed in Precam. The image shows a cut of the camera. The raft mounts below the window and over a copper plate that attaches to the coldhead

2 Pathfinder Instrument Overview

2.1 Focal Plane and Detector Architecture

The pathfinder focal plane is designed to demonstrate the performance of a moderately sized array of Skipper-CCDs operating in a survey configuration while preserving the key architectural elements required for future scaling to larger instruments. The focal plane consists of 20 Skipper-CCD sensors, each with a format of 1.35 Mpix, arranged to provide a contiguous imaging area corresponding to the optical field of the 0.6, m-class telescope at CTIO.

In the baseline configuration, four amplifiers per CCD are used, resulting in a total of 80 simultaneous video channels across the focal plane. This parallelization is a central feature of the design, allowing the system to achieve 2 second full-frame readout times while maintaining compatibility with Skipper-CCD multiple-sampling operation.

The readout architecture is based on the MIDNA ASIC, which provides low-noise analog front-end processing and supports both single-sample and multiple-sample readout modes. Groups of CCD outputs are multiplexed through the ASICs, reducing the number of downstream digitization channels while preserving signal integrity. The front-end electronics are located in close proximity to the detectors to minimize noise pickup and parasitic capacitance, while warm electronics outside the cryostat handle digitization, clock generation, and system control.

A key aspect of the focal plane design is the ability to operate the detectors in different readout modes depending on the science requirements. In fast survey mode, the system performs a single sample per pixel, achieving full-frame readout in approximately in 2 sec with 7e- readout noise. In contrast, in high-sensitivity mode, multiple non-destructive samples can be acquired for selected exposures, reducing the effective readout noise to the sub-electron level at the cost of increased readout time. This flexibility allows the instrument to probe a wide range of observational regimes within a unified hardware configuration.

The detectors are optimized for high quantum efficiency in the blue spectral range, with backside processing techniques under development to enhance sensitivity at wavelengths below 450 nm. The

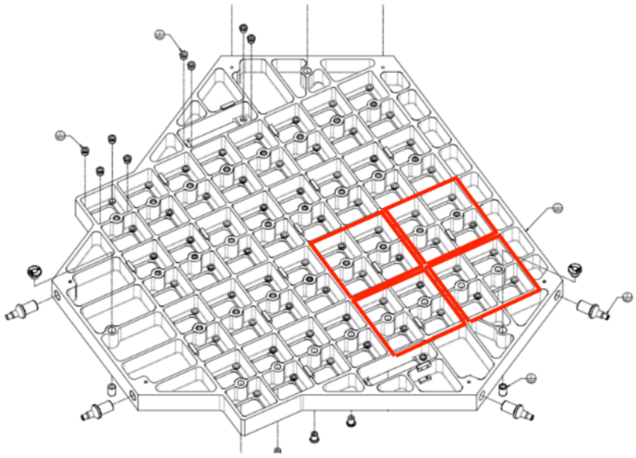

Figure 3. Image shows what would be four rafts in the ODI focal plane

sensor thickness and doping profiles are selected to ensure high charge collection efficiency and low dark current, while maintaining good response across the optical band.

Mechanically, the CCDs are packaged individually and integrated into a common focal plane structure that provides thermal stability, precise alignment, and reliable electrical connectivity. The focal plane assembly is coupled to the existing camera infrastructure, including vacuum vessel, cooling system, and optical interfaces, enabling rapid deployment while minimizing engineering overhead.

This architecture serves as a scalable prototype for future wide-field instruments based on large arrays of Skipper-CCDs. By validating the performance of a multi-amplifier, multiplexed readout system in an on-sky environment, the pathfinder establishes the foundation for extending this approach to focal planes with hundreds of sensors and thousands of readout channels.

A key design requirement of the pathfinder focal plane is compatibility with the existing One Degree Imager (ODI) at the WIYN 3.5 m telescope [12], enabling a direct path toward future deployment in a larger facility. To achieve this our pathfinder focal plane is a 20 CCD raft, that are designed to match the mechanical footprint of the ODI detector layout. In particular, the raft occupies the space of a 2×2 Orthogonal Transfer Array (OTA) CCD module in ODI. The pathfinder focal plane will be mounted on an adapter plate that reproduces the mechanical interface of the ODI focal plane, including mounting points, alignment references, and thermal interfaces. This approach allows the pathfinder to be installed within the ODI focal plane architecture with minimal modification to the existing instrument while also serving as a prototype for a full-scale replacement of OTA detectors with Skipper-CCD technology. Figure 3 shows what would be 4 out of 16 total rafts in ODI.

2.2 Readout and Noise Performance

The readout architecture is based on a highly parallelized implementation of the MIDNA ASIC (Figure 4) Each CCD is read out through its four output amplifiers, with approximately 3.38×10^5 pixels per amplifier. A single 16-channel MIDNA ASIC manages the front-end signal processing

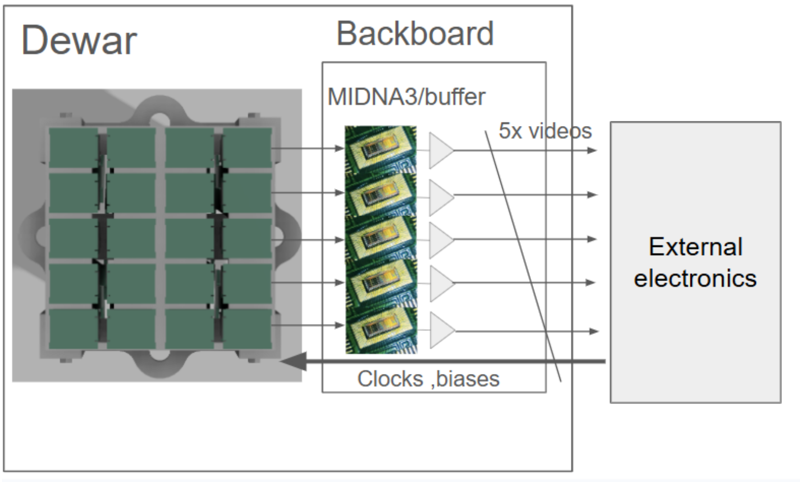

Figure 4. Readout electronics. The MIDNA chip perform the CDS and multiplexing of 16 channels, which are then amplified differentially and routed to the external electronics

for four separate detectors, providing features such as correlated double sampling and support for multiple non-destructive measurements.

Within the ASIC, all 16 channels are multiplexed into a single analog output channel, which in turn undergoes differential amplification through an external differential amplifier. This circuit is repeated five times, so the sample of all 20 skippers (80 video channels) produces only five differential video channels which are routed outside the Dewar for digitization and processing in the external controller electronics.

In the baseline fast-readout configuration, each pixel is sampled once with a readout time of $4\,\mu$s per pixel. This would result in a full-frame readout time of approximately 1.3 s per CCD, with a corresponding readout noise of $\sim 7\,e^-$ RMS. However, due to the multiplexing overhead time, we aim to have a readout time closer to 1.8 s. In addition, a low-noise operating mode is available in which each pixel is sampled multiple times. For example, with 50 samples per pixel, the effective readout noise is reduced to $\sim 1\,e^-$ RMS, with a full-frame readout time of approximately one and a half minutes. These two modes define the operational envelope of the system, enabling both high-cadence survey observations and higher-sensitivity measurements within the same hardware architecture.

2.3 Telescope and instrument (CTIO 0.6m)

The Curtis Schmidt telescope is located at the Cerro Tololo Interamerican Observatory (CTIO) in the Chilean Andes, specifically in the Coquimbo area (roughly 500 km north of Santiago), at an altitude of 2200 meters. Although functional, the telescope is currently inactive. It is owned by the University of Michigan and operated by the university's Department of Astronomy.

This telescope is a wide-field reflecting Schmidt telescope, featuring a 0.61-meter (24-inch) aperture. It has a 0.91-meter (36-inch) spherical primary mirror and an aspherical corrector plate. The telescope operates at a focal ratio of f/3.5. The optical design can be seen in figure 5

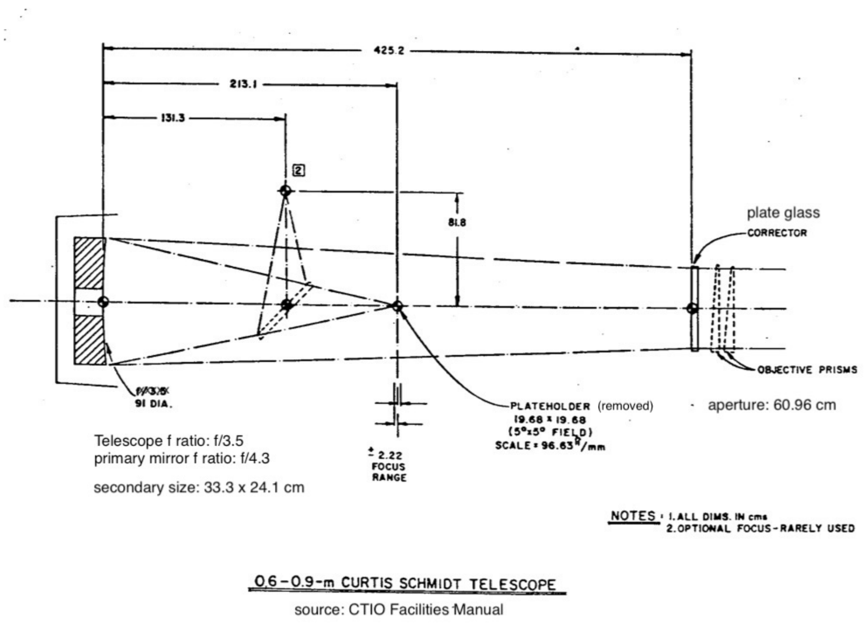

Figure 5. CS telescope optics, showig the newtonian design. The 45 degrees secondary mirror directs the light out through the side of the tube, creating a Newtonian-style focus where the camera is installed

The SkipperCam is mounted within the original Precam survey camera housing (model shown in Figures 6 and 2). This system utilizes a Polycold cryocooler, capable of reaching temperatures as low as approximately 120K.

The camera's original configuration contained two 2Kx4K DECam detectors. These have been removed and will be replaced by the detector raft detailed in Section 2.1

As illustrated in Figure 7, the camera system includes several components that will be upgraded with newer control electronics: a focus mechanism, a dark slide, a shutter, and a 5-position filter wheel. The shutter aperture and the filters are both 4-inches square (101.6 mm side), which is well-matched to the approximately 101x101 mm size of the new detector raft.

A redesign of the camera's front flange is planned to accommodate a 150 mm window and/or a field flattener. While final optical calculations are still pending, vignetting is anticipated due to the undersized secondary mirror. Replacing the secondary mirror would be a substantial effort, and this potential upgrade may be deferred to a later phase.

2.4 Survey Capabilities: Field of View, Cadence, Depth

The Curtis Schmidt telescope is a 0.61-meter telescope with an aperture of $f/3.5$, meaning that its focal length is 2135 mm. The proposed Skipper-CCD sensors have 1.35 Mpix distributed as 1278×1058 pixels of 15 μm in length each. For a normal lens focused at infinity the field of view can be calculated using the sensor size s and the focal length L_f as

$$\mathrm{FOV} = 2 \times \arctan\left(\frac{s}{2L_f}\right) \tag{2.1}$$

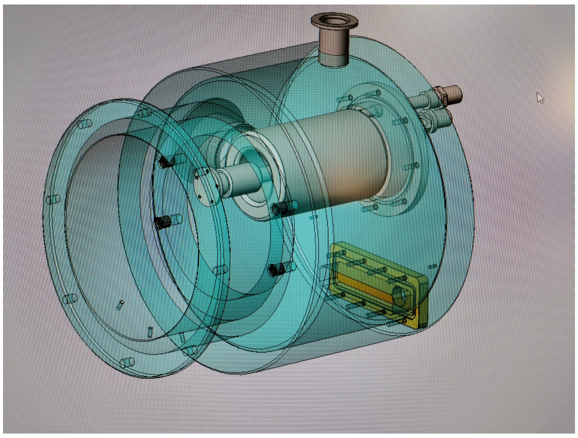

Figure 6. Skippercam (ex Precam) model, showing the cryocooler, vacuum port and hermetic for electrical signals

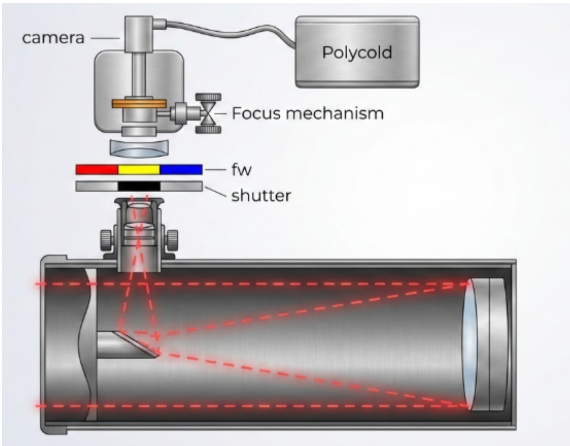

Figure 7. The light comes out of the side of the telescope, going through the shutter and filter wheel to get to the camera's window (and/or field flattener). The camera is mounted on a focusing mechanism

for the proposed survey this means a full FOV of 4.31 deg^2. Since the sensors will be placed leaving gaps between them, considering a 0.5 fill factor we get

$$\boxed{\mathrm{FoV}_{\mathrm{eff}} \approx 2.16\,\mathrm{deg}^2}$$

The optical depth of the Survey refers to the magnitude of the faintest source detectable at a given significance. In this case we will use threshold of 5σ to consider a detection as it is frequently used in astrophysical imaging. We will focus on the g-band filter (398-584) nm. Taking the PreCam exposure-time assumptions, the effective collecting area is taken to be 0.29 m^2. The pixel scale is 1.43"/pixel and the dark-sky background in g is $\sim$21 averaging over the whole month. For the following analysis we will consider a photometric calibration of $Z_p = 23.04$, based on the signal given by a zero magnitude source. The corresponding flux measured in the telescope due to the sky is then 6.3 e$^-$/s/pixel.

The PreCam model assumes that a photometric aperture of radius equal to one FWHM, taken to be 2", contains a fraction of 0.93 of the source flux. The corresponding aperture area is

$$N_{\rm pix} = \frac{\pi(2'')^2}{(1.43'')^2} \approx 6.15 \text{ pixels}, \tag{2.2}$$

since the sky flux is given per pixel then the expected background is

$$B = 6.3 \ \ \mathrm{e^-/s/pixel} \times 6.15 \text{ pixels} \times 2 \text{ s} = 77.49 \text{ e}^-. \tag{2.3}$$

Taking the 5σ threshold for detection and a 2 second exposure then the optical depth or the minimum flux can be obtained by solving the following equation:

$$S/N = \frac{S}{\sqrt{S+B}} \approx \frac{0.93 \times 2 \times f_{lim}}{\sqrt{0.93 \times 2 \times f_{lim} + 77.49 + 301.4}} = 5. \tag{2.4}$$

where a realistic readout noise (RN) of ~7 $\mathrm{e^-}$ contributing as $RN^2\mathrm{N}_{pix} = 301.4\ e^-$ to the total background. Converting this flux to magnitude, for a 2 s exposure, the limiting magnitude is approximately 18.7 and 20 for a 8 second exposure and a 2 s readout.

Another mode of observation for the proposed survey is by performing on-chip 3×3 binning, which reduces the number of pixels by a factor of nine and decreases the readout time from ~1.8 s to ~0.6 s. In this configuration, each binned pixel collects charge from nine physical pixels prior to readout, compressing the PSF of the device to approximately one pixel and increasing the effective pixel scale to 4.29"/pixel.

Using the assumptions described above, the expected sky background is

$$B = 6.3 \ \ \mathrm{e^-/s/pixels} \times 0.4 \text{ s} \times 9 \text{ pixels} = 23.4 \text{ e}^-. \tag{2.5}$$

Solving the signal to noise equation with a 5σ threshold, a readout noise of 7 $\mathrm{e^-}$ and $RN^2N_{pix} =$ 49 $\mathrm{e^-}$ (because the PSF compresses to one pixel) yields a limiting magnitude of 17.7. This mode enables very high cadence imaging at the $\sim$ 0.6, s level, which is well matched to the detection of fast transients and rapid variability. However, the short exposure time places the observations firmly in the read-noise dominated regime, resulting in a significantly shallower limiting magnitude compared to the nominal 2 s mode. This trade-off between cadence and depth highlights the flexibility of the instrument to probe different regions of time-domain parameter space.

3 Science Drivers

This work proposes an instrument that will serve two complementary purposes. First, it is a *pathfinder*: a scaled-down, on-sky demonstration of skipper-CCD technology as a next-generation detector for wide-field, high-cadence imaging, ahead of the full One Degree Imager (ODI-Skipper) planned for the WIYN 3.5 m telescope at Kitt Peak. Second, the pathfinder instrument is not merely an engineering testbed: its fast readout and blue sensitivity make it capable of an early, self-contained *science survey*, in the form of a high-cadence campaign sensitive to very short-timescale events. The remainder of this section describes both roles, the technology demonstration (Sec. 3.1) and the microlensing survey and its forecasts (Sec. 3.2).

3.1 One degree imager pathfinder

As has been described before, Skippercam is a small skipper-CCD instrument built around 20 sensors (1.35 Mpix each) and installed on the 0.6 m Curtis–Schmidt telescope at CTIO, reusing the existing PreCam camera body, shutter, and filter wheel. It delivers close to 2×2 deg field of view at a fast readout mode with a minimum ~2 s cadence with ~7 e^- readout noise. It realizes the initial phase of the full ODI-Skipper program extracting the maximum information from an on-sky technology demonstration while enabling early scientific results.

As a pathfinder, Skippercam will demonstrate several elements that are directly on the critical path for the full ODI-Skipper: the focal-plane mechanical and electronics design at raft scale, the MIDNA-based fast readout, the backside-processing solution that delivers the high blue (~350 nm) quantum efficiency central to the science case, and an early version of the ODI instrument-control and data-acquisition software, exercised on real multi-stream detector data. Successful on-sky operation of these elements retires the principal technical risks of the larger instrument while producing a usable scientific dataset.

3.2 Fast survey of the LMC with microlensing

The proposed Survey will monitor a single 2.16 deg^2 field centered on the LMC bar at a fixed cadence of $\Delta t = 10$ s during continuous ~ 8 hr nights, accumulated over 50 nights. With roughly $N_\star \simeq 1.35 \times 10^6$ source stars brighter than the survey limit $g < 20$ are monitored, drawn from the SMASH catalogue [15] from DEcam which uses similar filters. The primary method of observation will be microlensing of light curves. The high cadence is chosen to resolve the very short events (t_E of seconds to hours) expected from low-mass and compact lenses, giving sensitivity to planetary- and sub-planetary-mass primordial black holes and to fast-moving compact-object populations. The full instrumental, observing, and target parameters are collected in Table 1 of Appendix A.

Gravitational lensing can be used to observe non-luminous massive objects at astronomical distances. Light passing in the vicinity of a massive object is bent by the object's gravitational field, causing the light from background stars or sources to be distorted by massive objects or lenses that lie along the line of sight. How much the light is magnified and how long the event lasts depends on the mass of the lens M and its effective angular diameter θ_E given by

$$\theta_E = \sqrt{\frac{4GM(1 - d_L/d_s)}{c^2 d_L}} \tag{3.1}$$

where d_L and d_s are the lens and source distances respectively. The relationship between the angular diameter of the source θ_* and that of the lens θ_E separates microlensing in two distinct regimes. The $\theta_* << \theta_E$ regime is called the point-source regime, here the angular extent of the source is negligible, typical for large lens masses and distant sources. The duration of the event in this case is given by the time it takes for the source to cross the Einstein radius of the lens known as Einstein crossing time,

$$t_E = \frac{\theta_E}{\mu_{rel}} \tag{3.2}$$

with μ_{rel} the relative proper motion of the source and the lens. The magnification of the event is given by the impact parameter u following the formula,

$$A_{ps}(u) = \frac{u^2 + 2}{u\sqrt{u^2 + 4}} \tag{3.3}$$

that yields a light curve with a narrow peak when the microlensing event occurs with a duration of $\sim 2t_E$.

In the case where $\theta_* \geq \theta_E$ these formulas break down causing the light curve to saturate at a lower maximum magnification and changing the duration the event to that of the crossing of the finite angular extent of the source with a timescale of $\sim 2\theta_*/\mu_{rel}$. In this case the magnification is given by the integral over the source disk given in polar coordinates as,

$$A_{finite}(u, \rho) = \frac{1}{\pi\rho^2} \int_0^{\rho} dr \int_0^{2\pi} d\phi r A_{ps}\left(\sqrt{r^2 + u^2 - 2urcos(\phi)}\right), \tag{3.4}$$

where $\rho = \theta_*/\theta_E$. The origin is chosen such that the lens center is located at a distance u from the axis along $\phi = 0$. In Figure. 8 we show two events simulated as seen by Skippercam 10 s cadence and passed through the detection pipeline. The two share identical parameters and differ only in the normalized source radius ρ, isolating the finite-source effect. The left panel is a point-source event ($\rho = 0.01$), whose light curve follows the point-source magnification of Eq. 3.3; the right panel shows the same configuration with $\rho = 2$, where finite-source effects saturate the peak magnification and broaden the event.

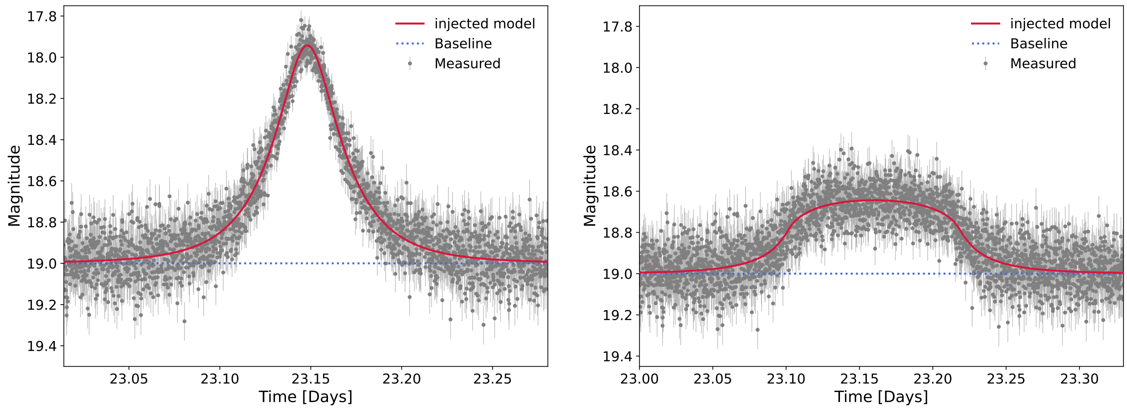

Figure 8. Two microlensing events simulated as seen by Skippercam's 10 s cadence and passed through the detection pipeline, sharing the same parameters ($t_E = 3 \times 10^3$ s, $u_0 = 0.4$, source $g = 19$) and differing only in the normalized source radius ρ. *Left:* a point-source event ($\rho = 0.01$), whose light curve follows the point-source magnification of Eq. 3.3. *Right:* the same configuration in the finite-source regime ($\rho = 2$), for which the peak magnification saturates and the event is broadened and flattened making it harder to detect. In both panels the points are the simulated photometry with their uncertainties and the red curve is the lensing magnification model.

The expected number of events from a generic microlensing survey is given by:

$$N_{exp} = T_{obs} \times N_\star \int_0^{\infty} \frac{d\Gamma}{dt_E} \epsilon(t_E) dt_E \tag{3.5}$$

where T_{obs} is the total observation time, $N_\star$ is the number of stars, $\frac{d\Gamma}{dt_E}$ is the differential event rate per star and the efficiency $\epsilon(t_E)$ relates to the probability of the survey's pipeline to detect an event of duration t_E.

3.2.1 Light Curve Injection and Noise Model

To estimate the detection efficiency of the survey we generate large ensembles of synthetic microlensing events and pass them through a detection pipeline that mimics the selection that would be applied to real data (Sec. 3.2.2). Each simulated event is sampled on an observing cadence of $\Delta t = 10$ for 50 continuous 8 hr nights.

For each injected event we draw a baseline source magnitude m_0 at random from the observed magnitude distribution of stars in the target field. The peak time t_0 is drawn uniformly across the observing window and the impact parameter u_0 uniformly in $[0, \max(1{,}2\rho)]$, corresponding to the threshold impact parameter $u_T = 1$ for a point source.

The light curve is computed first from the magnification of Eq. 3.3 in the point-source regime,

$$m(t) = m_0 - 2.5 \log_{10} A_{\rm ps}\big(u(t)\big), \tag{3.6}$$

and from the extended-source magnification of Eq. 3.4, $A_{\rm finite}(u, \rho)$, when finite-source effects are included (Sec. 3.2.3). The extended-source magnification is evaluated with the contour-integration scheme implemented in VBMicrolensing [16], pre-tabulated on a grid in $(\log u, \log \rho)$ and interpolated at run time for speed. This tabulation reproduces the point-source limit $A_{\rm finite}(u, \rho \to 0) = A_{\rm ps}(u)$ and recovers the expected suppression of the peak amplitude at large ρ.

Photometric noise is assigned time point using the survey's signal-to-noise relation. We convert the tabulated signal-to-noise ratio as a function of magnitude into a photometric uncertainty $\sigma_m = (2.5/\ln 10)/(S/N)$ and interpolate it as a function of the instantaneous (magnified) model magnitude, so that brighter measurements near the peak carry correspondingly smaller uncertainties. A more detailed treatment of noise due to source scintillations is reserved for further work.

3.2.2 Detection Pipeline

An injected event is counted as detected only if it passes a sequence of selection cuts applied to the simulated photometry. The cuts are inspired by the selection criteria of the OGLE high-cadence analysis [17], adapted to a clean simulation: criteria whose purpose in the real survey is the rejection of astrophysical or instrumental false positives (variable stars blending artifacts, multiple-bump light curves) are not reproduced here, since the simulated sample contains only genuine microlensing events. The cuts we retain are those that test whether a true event is statistically recoverable.

We first estimate the baseline level as the median of the light curve and define the per-time-point significance of brightening,

$$\chi_i = \frac{m_{\rm base} - m_{\rm obs}(t_i)}{\sigma_m(t_i)}, \tag{3.7}$$

positive for points brighter than baseline. A detection is triggered if *either* of two conditions is met: at least five points with $\chi_i > 3$, allowing a single isolated sub-threshold point within the run so that photometric noise does not artificially break an otherwise significant excursion; *or* at least three strictly consecutive points with $\chi_i > 7$. The second branch recovers very short but well-defined events, which may have too few epochs to satisfy the five-point condition yet produce individually high-significance points; requiring those three points to be strictly consecutive, together with the high per-point threshold, prevents scattered high noise from triggering. The longest qualifying run defines the "bump" used by the subsequent cuts.

We further require the peak significance within the bump to exceed 3σ, removing low-amplitude excursions that clear the per-point threshold only marginally. Casting this amplitude requirement in terms of significance rather than a fixed magnitude threshold is essential in our regime: the bright, low-noise sources that dominate the survey produce confidently detectable events even at small magnitude amplitudes, which a fixed magnitude cut would incorrectly reject.

We also require the photometry outside the bump to be consistent with a flat baseline. Provided enough epochs lie outside the bump, we compute the reduced chi-square of the out-of-bump residuals about the source baseline and reject the event if it exceeds a threshold set at 4. In the clean simulation this removes light curves whose flux varies significantly away from the identified bump.

We then fit the microlensing model to the light curve by weighted least squares, with the baseline magnitude, peak time t_0, impact parameter u_0, timescale t_E and, in the finite-source analysis, the normalized source radius ρ as free parameters. The fit is performed on a window spanning the measured bump (with padding), and the peak time is initialized at the bump centre. We require the reduced chi-square to satisfy $\chi^2/\text{dof} \le 2$ over the full fit window, and $\chi^2/\text{dof} \le 4$ over the epochs within $|t - t_0| < t_E$ and within $|t - t_0| < 2t_E$; the near-peak ranges are tested only when they contain enough points to form a meaningful reduced chi-square, so that short, sparsely sampled events are not rejected on a statistic that cannot be evaluated. Requiring a good fit both globally and near the peak enforces that the recovered light curve has the microlensing shape.

The detection efficiency is the fraction of injected events passing all of the above. For the point-source baseline it is a function of event duration alone, $\epsilon(t_{\text{dur}})$; for the finite-source analysis it is a two-dimensional function $\epsilon(t_{\text{dur}}, \rho)$ of duration and normalized source radius. For each cell of the grid we inject $N = 2500$ events, drawing m_0, t_0 as described previously, and tally the fraction detected. The impact parameter is sampled uniformly in $u_0 \in [0, \max(1, 2\rho)]$, so that grazing events whose peak magnification falls below threshold in the finite-source regime are correctly counted as losses rather than excluded from the sample; the resulting efficiency is a normalized detection probability for each (t_{dur}, ρ).

The resulting point-source curve exhibits three features, each set by a distinct timescale of the survey. At short durations the efficiency rises from zero near $t_{\text{dur}} \sim$ a few $\times\,\Delta t$: events shorter than the sampling cannot produce the required run of significant points. At intermediate durations the efficiency reaches a plateau where events are well sampled and are recovered with high probability once above threshold. At long durations the efficiency falls again as the event duration approaches the length of the observing window, beyond which the event no longer rises and falls within the data and the timescale can no longer be constrained. The location of the short-duration turn-on is therefore set by the cadence and that of the long-duration cutoff by the observing baseline, defining the range $[t_{E,\min}, t_{E,\max}]$ over which $\epsilon > 0$. Combining multiple nights extends the efficiency to longer timescales and thus to higher lens masses.

3.2.3 Finite-Source Treatment

For finite sources, the simulation is run on a grid of ρ spanning $\log_{10}\rho \in [-3, 1]$. The extended-source magnification is evaluated with the tabulated VBMicrolensing scheme that solves Eq. 3.4. The resulting matrix reduces to the point-source efficiency in the limit $\rho \to 0$ and is strongly suppressed for $\rho \gtrsim 1$, vanishing once the saturated peak amplitude drops below the detection threshold.

The normalized source radius depends on both the lens (through θ_E) and the source (through θ_*). For each lens mass M_L and line-of-sight position $x = d_L/d_s$, θ_E follows from Eq. 3.2. The source angular size is set by the apparent magnitude of the source through the surface-brightness relation, which for the target field we take in the form [17]

$$\theta_* = 0.91\ \mu\text{as} \times 10^{-0.2\,(I_s - I_{\rm RC})}, \tag{3.8}$$

where I_s is the source I-band magnitude and $I_{\rm RC} = 18.5$ is the red-clump reference magnitude of the field. Although the survey selects sources in g, we evaluate θ_* from the I-band magnitude of the same stars (drawn also from the SMASH catalogue).

A real field contains sources of a range of magnitudes, and hence a range of physical sizes, so the efficiency entering the rate must be averaged over the source population. Denoting by $\phi(I_s)$ the normalized I-band magnitude distribution of the detectable ($g < 20$) sources, the effective efficiency at fixed (t_E, x) is

$$\bar{\epsilon}(t_E, x) = \int dI_s\ \phi(I_s)\ \epsilon\Big(t_E,\ \rho\big(I_s, x\big)\Big), \tag{3.9}$$

with $\rho(I_s, x) = \theta_*(I_s)/\theta_E(x)$. This $\bar{\epsilon}$ replaces $\epsilon(t_E)$ in the rate integral Eq. 3.13, so that the finite-source wall is set self-consistently by the actual source-size distribution of the field. Because the source distribution is dominated by red-clump giants near $I_s \simeq 18.5$, the wall is driven by these comparatively large sources, with the fainter (smaller) tail of the population extending the reach modestly to lower masses.

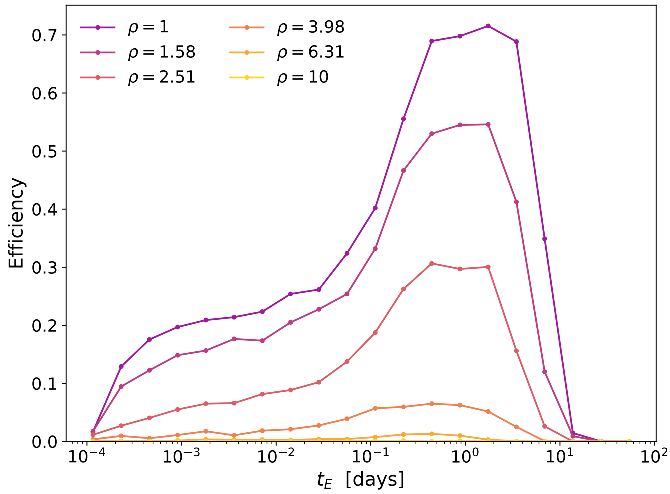

Figure 9. Detection efficiency $\varepsilon(t_E, \rho)$ of the proposed pipeline as a function of the Einstein-radius crossing time t_E, for several values of the normalized source radius $\rho = \theta_*/\theta_{\rm E}$ (colored curves). At fixed ρ the efficiency rises at short t_E as events become long enough to be sampled by several epochs, plateaus where the event is well resolved, and falls again at the longest t_E as events approach the survey duration and no longer return to baseline. The strong suppression with increasing ρ is the finite-source effect: for $\rho \gtrsim 2.5$ the magnification of a source larger than the Einstein ring is washed out, and by $\rho = 10$ the efficiency is negligible at all t_E.

3.3 Fast lenses

As mentioned before, finite-source effects limit the sensitivity of a high-cadence survey at lower masses. The degeneracy between lens mass and velocity in the microlensing relations provides a workaround which has recently been put forward by [18]: at a fixed event duration t_E, a faster lens corresponds to a more massive one, so a high-velocity population produces short-duration events at masses well above those accessible to a slow population. For conventional dark matter searches and free-floating planets this offers no relief, since the lens velocity is effectively fixed to the virialized halo expectation, $v \sim 220$ km/s (or the lower internal dispersion of the host system for FFPs).

A number of theoretical scenarios, however, predict populations of compact objects whose velocities depart dramatically from the virialized-halo value, and which would therefore populate qualitatively new regions of the mass–timescale plane. As highlighted by [18], these include: ultrarelativistic primordial black holes produced in the collapse of cosmic strings; mirror neutron stars arising in dark-sector models; black-hole merger remnants that receive large gravitational recoil ("kick") velocities; dark-matter nuggets; free floaters ejected from gravitationally bound systems; and compact objects formed in the disk rather than the halo. The transverse speeds of such populations can range from a modest enhancement over the halo value up to a substantial fraction of the speed of light.

As a benchmark we adopt the fast lens population moving at $v_0 = 0.1c$, following [18]; this is representative of the relativistic end of the scenarios above, while the same framework applies at any chosen v_0.

3.3.1 Model-Independent Rate Limits

We first derive limits on the microlensing event rate that are independent of any assumption about the lens mass or velocity distribution. Following the standard construction [17, 19], the 95% confidence-level upper limit on the event rate per source star, as a function of the Einstein crossing time, is

$$\Gamma_{\rm excl}(t_E) = \frac{N_{\rm excl}}{N_\star\, T_{\rm obs}\, \epsilon(t_E)}, \tag{3.10}$$

where $N_\star$ is the number of monitored source stars, $T_{\rm obs}$ the total observing time, $\epsilon(t_E)$ the detection efficiency of Sec. 3.2.2, and $N_{\rm excl}$ the number of events excluded at 95% C.L. for the number actually observed. For a survey with no detected events we adopt the Poisson upper limit $N_{\rm excl} = 3.0$. It is worth mentioning that the timescale used in our light-curve model, Eq. 3.2, is the Einstein *radius* crossing time $t_E = \theta_E/\mu_{\rm rel}$, whereas the limits in the comparison surveys [17, 19] are quoted in terms of the Einstein *diameter* crossing time, larger by a factor of two. We convert our timescales accordingly, $t_E^{\rm diam} = 2\,t_E$, before placing our curve on the common axis. The results are shown in Figure. 10

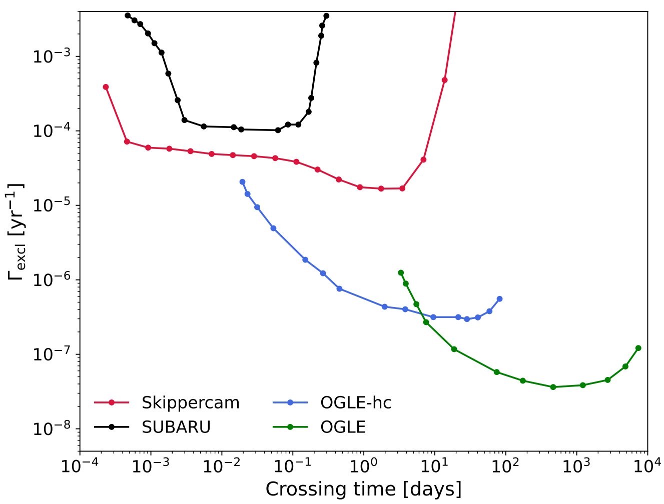

Figure 10. Model-independent 95% confidence-level upper limit on the microlensing event rate per source star, $\Gamma_{\rm excl}$, as a function of the event crossing time, for Skippercam (red) compared with the published Subaru-HSC (black), OGLE-hc (blue), and OGLE (green) surveys. The limit is obtained from Eq. 3.5 assuming no detected events, and depends only on the number of monitored sources, the observing time, and the detection efficiency, making no assumption about the lens population.

3.3.2 Mass-Dependent Limits

To turn the model-independent rate limit into a constraint on a specific lens population, we evaluate the expected number of events, Eq. 3.5, for a monochromatic mass function $f(M) = \delta(M - M_L)$ and present the result as a function of M_L as is usual in microlensing surveys. This requires assumptions about the lens kinematics and spatial distribution. For the velocity we take a Maxwell–Boltzmann distribution with one-dimensional dispersion v_0, whose projected speed distribution is $f_\perp(v_\perp) \propto v_\perp \exp(-v_\perp^2/v_0^2)$. Following Ref. [18] our benchmark is a fast-lens population with

$v_0 = 0.1c$, representative of compact objects far from the virialized-halo velocity. The framework applies to any v_0, but the resulting mass mapping is benchmark-specific.

The spatial distribution will be tested under two assumptions. The lenses may be distributed *uniformly* along the line of sight, $\rho_M(d_L) = \tilde{r}\,\rho_{\rm DM}^{\odot}$ with $\rho_{\rm DM}^{\odot} = 0.3\ {\rm GeV\,cm^{-3}}$, as expected for a population that does not cluster like the halo; or they may *trace the dark matter*, following the Navarro–Frenk–White (NFW) profile

$$\rho_{\rm NFW}(r) = \frac{\rho_0}{(r/r_s)\,(1 + r/r_s)^2}, \tag{3.11}$$

summed over the Milky Way and LMC haloes, $\rho_{\rm NFW}^{\rm tot}(x) = \rho_{\rm NFW}^{\rm MW}(x) + \rho_{\rm NFW}^{\rm LMC}(x)$, with the characteristic densities and scale radii of Ref. [18], so that $\rho_M(x) = \tilde{r}\,\rho_{\rm NFW}^{\rm tot}(x)$. In both cases $\tilde{r}$ is the fraction of the relevant density carried by the fast-lens population, and the two assumptions span the agnostic and halo-tracing extremes.

The survey's reach to low masses follows from a degeneracy in the microlensing relations: at fixed event duration, lens mass and transverse velocity trade off through the Einstein crossing time. Writing $x = d_L/d_s$ and combining $t_E = 2u_T\theta_E d_s/v_\perp$ with Eq. 3.2,

$$v_\perp(x) = \frac{2u_T}{t_E}\sqrt{\frac{4GM_L}{c^2}\,x(1-x)\,d_s}. \tag{3.12}$$

At fixed v_0 a shorter t_E maps to a lower mass, which is how the high-cadence (short t_E) sensitivity translates into reach at low M_L. Conversely, at fixed t_E a faster lens corresponds to a higher mass.

Substituting the speed distribution into the differential rate of Eq. 3.5, the expected number of events for either spatial assumption is

$$N_{\rm ev}(M_L) = N_\star T_{\rm obs}\int_0^1 dx \int dt_E\ \epsilon(t_E,\rho)\,\frac{2d_s}{v_0^2 M_L}\,\rho_M(x)\,v_\perp^4(x)\,e^{-v_\perp^2(x)/v_0^2}, \tag{3.13}$$

with $v_\perp(x)$ as above, $u_T = 1$ for a point source, and the efficiency evaluated at the model (radius) crossing time corresponding to each (t_E, x). Because the rate is linear in the lens density, $N_{\rm ev} \propto \tilde{r}$, we evaluate Eq. 3.13 once at $\tilde{r} = 1$ and read off the 95% C.L. upper limit directly,

$$\tilde{r}_{\rm lim}(M_L) = \frac{N_{\rm excl}}{N_{\rm ev}(M_L;\,\tilde{r}=1)}. \tag{3.14}$$

For the uniform case $\tilde{r}_{\rm lim}$ is the limiting density in units of the local dark matter density, for the NFW case it is the limiting fraction of the halo density carried by the population.

Figure 11 shows the resulting limits for both spatial assumptions, alongside the published OGLE [20], OGLE-hc [17], and Subaru-HSC [19] constraints. The high-cadence reach extends the sensitivity to low lens masses, where it is ultimately cut off by the shallow finite-source wall (Sec. 3.2.3). The Subaru-HSC curve is drawn dashed because a recent reanalysis of those data [21] has revised the original constraints.

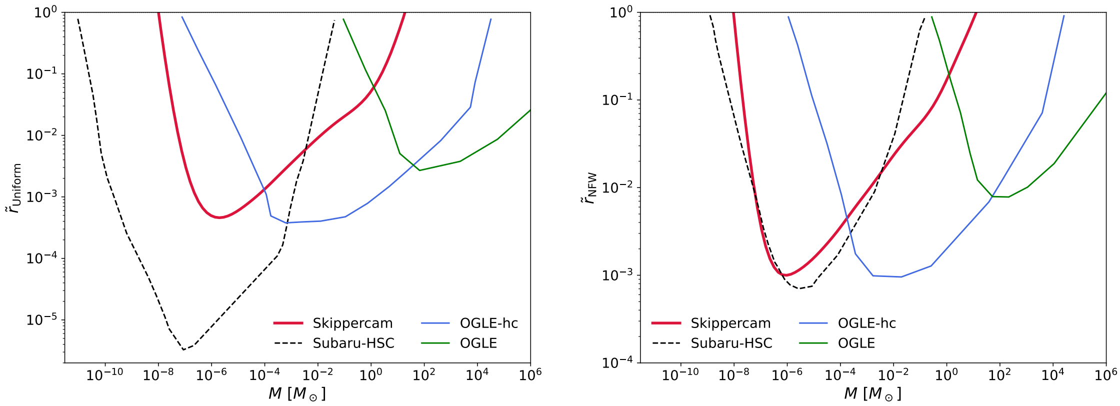

Figure 11. Model-dependent 95% C.L. upper limits on the fast-lens population ($v_0 = 0.1c$) as a function of monochromatic lens mass M_L. *Left:* lenses distributed uniformly along the line of sight, normalised to the local dark matter density $\rho_{\rm DM}^{\odot}$. *Right:* lenses tracing the summed Milky Way + LMC NFW halo, normalised to the local halo density. The Skippercam limit (this work) is shown with the published OGLE, OGLE-hc, and Subaru-HSC constraints; the Subaru-HSC curve is dashed to reflect the recent reanalysis [21].

3.3.3 Primordial Black Holes at the LMC

The fast lens benchmark of Sec. 3.3.2 kept the velocity a free parameter. We now specialize to the physically motivated case of primordial black holes (PBHs) that make up a fraction $f = \Omega_{\rm PBH}/\Omega_{\rm DM}$ of the dark matter and are virialized in the Milky Way and LMC haloes, so their kinematics are fixed rather than chosen. Lens velocities are drawn from a Maxwellian with one-dimensional dispersion $\sigma_v \simeq V_c/\sqrt{2} \approx 155\ {\rm km\,s^{-1}}$ in the Galactic rest frame, to which we add the projected solar motion and the proper motion and internal dispersion of the LMC sources. The relative proper motion $\mu_{\rm rel} = |\boldsymbol{\mu}_L - \boldsymbol{\mu}_s|$ then sets the crossing time $t_E = \theta_E/\mu_{\rm rel}$ (Eq. 3.2) for each event. As a check on the geometry and velocity treatment, our event generator reproduces the familiar rate-weighted scaling $\langle t_E \rangle \simeq 62\sqrt{M/M_\odot}$ d [17].

Two density components contribute along the line of sight. The Milky Way halo is a contracted NFW profile normalized to the local density $\rho_{\rm DM}^{\odot} = 0.3\ {\rm GeV\,cm^{-3}}$ [22], with the limits depending only weakly on the assumed profile [17]. The LMC self-halo is a Hernquist profile of total mass $1.49 \times 10^{11}\ M_\odot$ and scale length 17.1 kpc [17, 23], evaluated at the LMC-centric radius along the line of sight. For each component the lens number density is $n_k(x) = \rho_{{\rm halo},k}(x)/M$, so the rate is linear in f.

Folding the source-averaged finite-source efficiency of Eq. 3.9 into the rate, the expected number of detected events for a PBH fraction f is

$$N_{\rm exp}(f, M) = f\, N_\star\, T_{\rm obs} \sum_{k\in\{\rm MW,\, LMC\}} \int dt_E \int d\rho\, \frac{d^2\Gamma_k}{dt_E\, d\rho}\, \epsilon(t_E, \rho), \tag{3.15}$$

where each component's differential rate $d^2\Gamma_k/dt_E\, d\rho$ is built by Monte-Carlo event generation [24] following [20], in each trial we draw the lens distance d_L weighted by $\rho_{{\rm halo},k}$, sample the lens and source velocities, and assign a source magnitude I_s from the field luminosity function. The angular source size θ_* then follows from Eq. 3.8 and the normalized radius from $\rho = \theta_*/\theta_E$, so drawing I_s

per trial realizes the source average of Eq. 3.9 event by event. Since $N_{\rm exp} \propto f$, the 95% C.L. upper limit for a null detection ($N_{\rm excl} = 3$) follows directly,

$$f_{95}(M) = \frac{N_{\rm excl}}{N_{\rm exp}(f{=}1, M)}. \tag{3.16}$$

The resulting limit $f_{95}(M)$, shown in Figure 12, has the shape characteristic of a finite-source-limited constraint. The low-mass wall is set by finite-source suppression: as M decreases, θ_E shrinks, ρ grows, and the peak magnification saturates below the detection threshold. Because the bright ($g < 20$) source sample is dominated by large red-clump giants, this wall sits at higher masses than for deeper surveys an effect we examine in detail in Sec. 3.2.3. We compute the limit here with the same methodology as OGLE [20] so that it is directly comparable with the other surveys; as a consistency check, feeding the halo velocity ($v_0 = 7.34 \times 10^{-4}$ c) into the general fast lens framework of Sec. 3.3.2 reproduces this same limit.

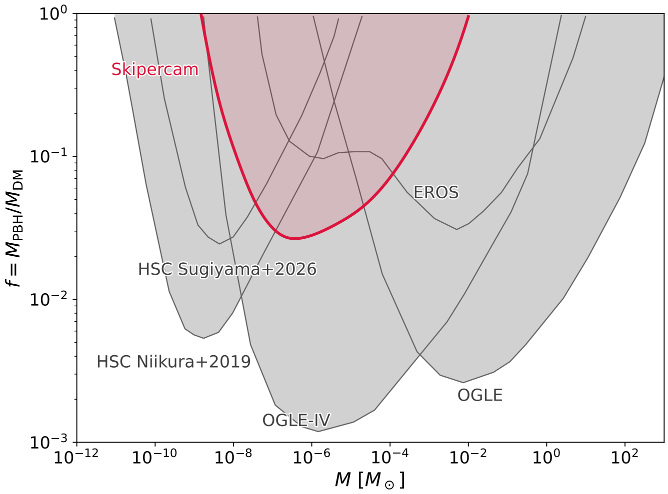

Figure 12. Projected 95% C.L. upper limit on the fraction of dark matter in primordial black holes, $f = M_{\rm PBH}/M_{\rm DM}$, as a function of PBH mass for Skippercam (red). Shaded grey regions show existing constraints from EROS [25], OGLE [20], OGLE-IV [17], and Subaru-HSC [19, 21].

3.4 Finite-source suppression and the magnitude limit

As we have sen microlensing event is suppressed by finite-source effects when the angular size of the source, θ_*, becomes comparable to or larger than the Einstein radius $\theta_{\rm E}$, i.e. when the normalized source radius $\rho \equiv \theta_*/\theta_{\rm E}$ exceeds unity. For $\rho \gtrsim 2.5$ the peak magnification is reduced to the point where the event very hard to detect in practice as can be seen from Figure. 9. Because $\theta_{\rm E} \propto \sqrt{M}$, this wall moves to ever larger ρ as the lens mass decreases, so any survey targeting sub-planetary lenses is pushed towards the finite-source regime. What sets Skippercam apart from deeper surveys is *not* this mass scaling, which is common to all of them, but the angular size of the sources it is able to monitor.

Skippercam detects sources down to a limiting magnitude of $g < 20$. As Figure 13 (left) shows, this limit removes the bulk of the LMC stellar population, which extends several magnitudes fainter and retains only the intrinsically luminous stars. Since luminosity correlates with physical radius at the (essentially fixed) distance of the LMC, the surviving sources are systematically the *largest* on the sky: their characteristic angular size θ_* is correspondingly large. A survey reaching several magnitudes deeper would instead be dominated by fainter, more compact sources with smaller θ_*.

The consequence for finite-source suppression is shown in Figure 13 (right), where we evaluate ρ for a representative low-mass lens, $M = 10^{-7}\,M_\odot$. The full source population (blue) peaks near $\rho \simeq 2$, straddling the finite-source threshold, whereas the $g < 20$ sample that Skippercam actually observes (red) is shifted almost entirely to $\rho > 2.5$. In other words, for the low-mass lenses where Skippercam's high cadence would otherwise grant access, its bright, large-θ_* sources sit predominantly on the wrong side of the finite-source wall. Deeper surveys probing the same lens masses populate the $\rho \lesssim 1$ regime and are therefore far less affected by finite-source suppression at fixed M.

This is a structural limitation of a shallow, bright-source-limited survey rather than a deficiency of the detection pipeline: the same magnitude limit that makes the targets easy to monitor at high cadence also guarantees that they are large on the sky, and hence maximally vulnerable to finite-source effects in exactly the low-mass regime of interest. It is the dominant reason Skippercam is more strongly affected by finite-source suppression than deeper microlensing surveys such as OGLE or Subaru-HSC at comparable lens masses.

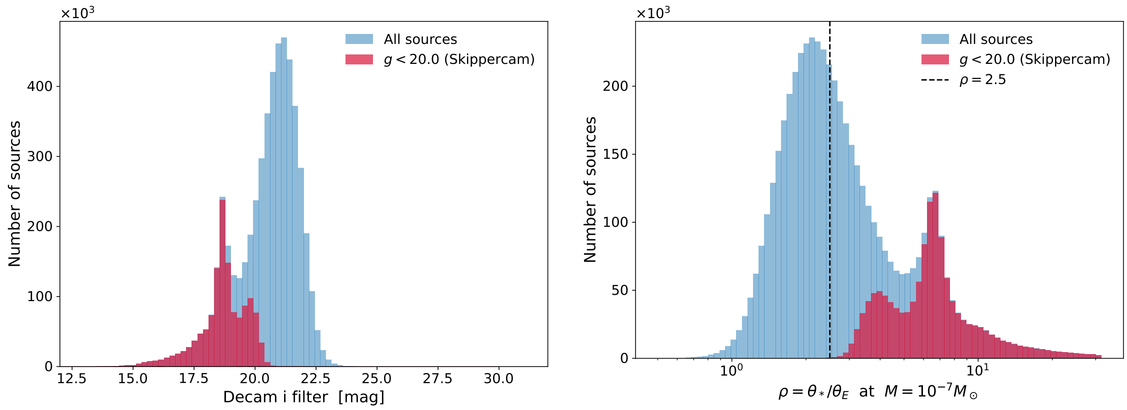

Figure 13. The effect of Skippercam's shallow magnitude limit on its susceptibility to finite-source suppression. *Left:* distribution of LMC source stars in the native DECam i band. The full catalogue (blue) extends to $i \simeq 30$, but Skippercam's detection limit of $g < 20$ (red) retains only the intrinsically bright end of the population. *Right:* the resulting distribution of the finite-source parameter $\rho = \theta_*/\theta_\mathrm{E}$ for a lens of mass $M = 10^{-7}\,M_\odot$. Brighter sources are physically larger (larger θ_*), so the magnitude-limited sample (red) is displaced to systematically higher ρ than the full population (blue), piling up almost entirely beyond the finite-source threshold $\rho \simeq 2.5$ (dashed line) above which the magnification is strongly suppressed.

3.4.1 Impact on Microlensing Systems Characterization

Wide-area microlensing surveys discover large numbers of events but often at a cadence insufficient to verify their lensing nature or to accurately recover the physical parameters of the lens system. This limitation has motivated dedicated higher-cadence efforts: the OMEGA Key Project, for example, uses the Las Cumbres Observatory network to follow up survey-detected candidates with continuous, multi-aperture monitoring, precisely because the parent surveys' cadence is too sparse to constrain the lens masses and distances [26]. Skippercam can fill this complementary role in a different mode: rather than triggering follow-up of individual alerts, its high-cadence, wide-field staring strategy continuously samples every event in its field on second timescales, delivering densely sampled light curves for the full population without the need for external triggering.

This fast-cadence coverage is valuable because different regions of a microlensing light curve encode information about distinct model parameters (Figure. 14): finite-source effects are constrained primarily by the morphology of the peak, parallax signatures by long-timescale asymmetries and inter-observatory offsets, and planetary or companion parameters by localized deviations associated with caustic perturbations. Dense, rapid sampling of the peak is particularly important for resolving finite-source and short-timescale features, which are washed out at the cadences typical of wide-area surveys. In combination with such surveys, Skippercam therefore enables the detection of shorter-timescale events, improves the measurement of higher-order effects such as parallax and finite-source signatures, and increases the number of events with well-measured lens parameters. The pairing of broad statistical samples from wide-area surveys with high-cadence characterization opens a regime in which large samples and precise per-event parameter measurements can be achieved together, ultimately yielding a more comprehensive understanding of the Galactic and LMC halo lens populations.

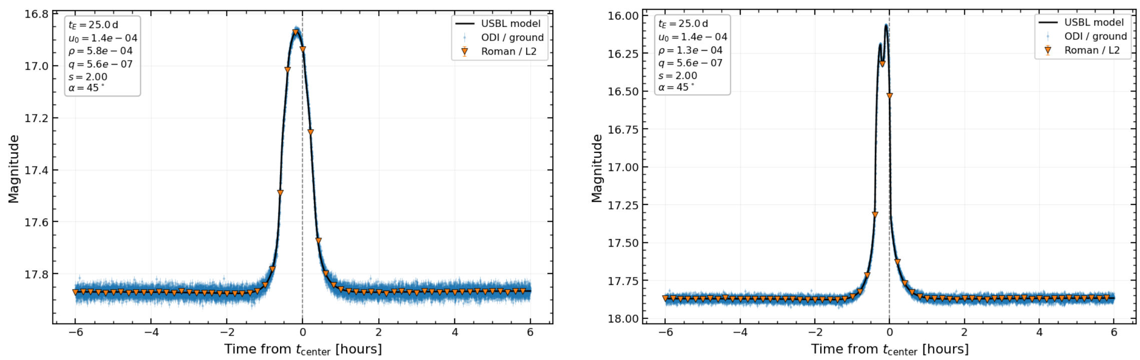

Figure 14. Examples of short-timescale anomalies in a binary-lens event produced by a planetary system composed of a $10\,M_\odot$ host star and a $2\,M_\oplus$ planet. The figure illustrates a case in which high-cadence SkipperCam observations (assumed to be 5 seconds) can improve the characterization of a planetary anomaly that is only sparsely sampled by the Nancy Grace Roman Space Telescope ($\sim$ 15 min cadence). The photometric bands have been artificially aligned for illustrative purposes.

3.5 LSST-Driven Stellar Occultations as a Prime Science Case for SkipperCam

The Legacy Survey of Space and Time (LSST) [9] at the Vera C. Rubin Observatory will transform Solar System science by discovering and characterizing an unprecedented population of small bodies. It is expected to detect tens of thousands of trans-Neptunian objects (TNOs), along with a large number of main-belt asteroids and near-Earth objects. Combined with Gaia astrometry, LSST will enable highly accurate ephemerides and a dramatic increase in the number of predictable stellar occultations [27]. The combination of LSST astrometry and Gaia star catalogs enables predictions of stellar occultations with kilometer-scale accuracy on the Earth's surface.

Stellar occultations provide a uniquely powerful method for measuring the sizes, shapes, and environments of these objects, often with kilometer-scale precision that rivals spacecraft observations [28]. The key observable is the time-resolved light curve of a star as it is occulted, converting temporal resolution into spatial resolution on the object.

The scientific return of LSST-discovered objects will therefore critically depend on the ability to follow up predicted occultations with high temporal resolution and high sensitivity. However, fully exploiting LSST-enabled occultations exposes a fundamental limitation of conventional detectors: the inability to achieve both high temporal resolution and high sensitivity simultaneously.

This limitation becomes particularly acute in the LSST era for three reasons:

1. **Fainter targets:** LSST will probe deeper populations, leading to occultations involving fainter stars and smaller bodies, requiring higher sensitivity at short exposure times.

2. **Higher event rates:** The number of predicted occultations will increase dramatically, demanding efficient, high-cadence follow-up capabilities.

3. **Smaller-scale features:** Key observables such as Fresnel diffraction and projected stellar diameters occur on kilometer or sub-kilometer scales, requiring sub-second cadence to resolve.

For typical shadow velocities of $V \sim 10$–$30\ \mathrm{km\,s^{-1}}$, a 1-second exposure corresponds to spatial smearing of 10–30 km, significantly larger than the Fresnel scale (~ 1 km for TNO distances) [28]. Thus, fully exploiting LSST-enabled occultations requires cadences at the ~ 0.1 s level or faster, while maintaining sufficient S/N.

In the read-noise-limited regime relevant for sub-second exposures, reducing the read noise from ~ 5–$10\ e^-$ to $\lesssim 1\ e^-$ results in a factor of several improvement in signal-to-noise ratio. SkipperCam directly addresses this challenge. Its combination of sub-electron readout noise and flexible readout modes enables high signal-to-noise photometry at short exposure times, effectively breaking the traditional trade-off between cadence and sensitivity. This capability uniquely positions SkipperCam as a follow-up instrument for LSST-driven occultation science.

In this context, SkipperCam enables:

- high-precision timing of occultation events with sub-exposure accuracy,
- sensitivity to faint stars and small ($\lesssim$ km-scale) occulting bodies,
- resolution of diffraction patterns and stellar angular sizes,
- detection of tenuous atmospheres, rings, and close companions,
- and efficient follow-up of the large volume of LSST-predicted events.

This capability is particularly relevant for probing the size distribution of kilometer-scale objects in the outer Solar System, a key observable for understanding planetesimal formation and collisional evolution.

In this context, LSST provides the discovery and prediction capability, while SkipperCam provides the measurement capability. Together, they enable a new regime of occultation science that is currently inaccessible, combining faint targets with sub-second temporal resolution.

4 Data Processing and Analysis

The proposed survey obtains continuous, high-cadence imaging of a single 2.16 deg^2 field in the LMC at a cadence of 10 s across the 20 skipper-CCDs of the focal plane. Over an 8-hour night this yields ~2880 exposures per sensor, or ~57600 individual sensor images. At ~2.6 MB per image this corresponds to a raw data volume of ~0.15 TB per night, and ~7.5 TB over the full 50-night campaign a manageable rate that nonetheless places demands on the storage and processing infrastructure.

4.1 Data management and processing strategy

The data rate, while modest compared to facilities such as the Vera C. Rubin Observatory (~20 TB per night), is large enough that storing and reprocessing every raw frame indefinitely is neither necessary nor efficient. We will adopt the standard strategy of large imaging surveys (DECam, Rubin/LSST): raw frames are reduced promptly on a mountain-top compute cluster co-located with the instrument, the science-ready data products (calibrated images and, principally, the photometric catalogs and light curves) are retained, and the bulk raw data are archived rather than kept on disk. Because the scientific product of a high-cadence microlensing survey is the *light curve* of each source rather than the individual images, the long-term data footprint is dominated by the compact catalog-level products, which are orders of magnitude smaller than the raw image stream.

4.2 Image reduction and photometry

Each raw frame undergoes standard CCD reduction prior to analysis. Master bias frames, constructed from a nightly stack of zero-second exposures, are subtracted to remove the electronic offset introduced during readout. Master flat fields, derived from twilight sky exposures, are then applied to correct for pixel-to-pixel sensitivity variations and large-scale illumination gradients across the focal plane. Given the high stellar density of the LMC field, accurate astrometric registration of every frame to a common reference is required so that the same physical source is tracked across all epochs.

Photometry is performed in the crowded-field regime characteristic of the LMC. Source detection and deblending with `SExtractor` [29] identifies contiguous pixel clusters above a signal-to-noise threshold; however, in the densely populated inner LMC, aperture photometry alone is degraded by blending, and point-spread-function (PSF) fitting photometry (e.g. `DAOPHOT`-style, or difference-image photometry relative to a co-added template) is used to recover accurate fluxes for individual stars. Source catalogs are cross-matched across epochs by celestial coordinates to assemble a photometric time series for each object spanning the full night and, across nights, the full campaign. The final data product will be a catalog of light curves, one per detected source. These light curves are the input to the microlensing event-detection pipeline described in Sec. 3.2, which searches for the characteristic transient magnification of a background star.

5 Conclusions

We have presented the science program and observational strategy for SkipperCam, a Skipper-CCD imager pathfinder built around a 20-sensor focal plane (~27 Mpix) installed on the 0.6 m Curtis–Schmidt telescope at CTIO, reusing the existing PreCam camera body, shutter, and filter wheel. By coupling Skipper-CCD detectors to a massively parallel, MIDNA-based readout architecture, the instrument decouples the traditional trade-off between low noise and slow readout at the system level: it delivers a wide $\sim 2\,\mathrm{deg}^2$ field of view with ~2 s full-frame cadence and $\sim 7\,e^-$ readout noise in fast survey mode, while retaining the ability to reach sub-electron noise through multiple non-destructive sampling. This combination opens a new observational regime in which wide-field, high-cadence imaging becomes photon-limited rather than read-noise-limited.

The core science driver is a high-cadence microlensing survey of the Large Magellanic Cloud, monitoring a single $2.16\,\mathrm{deg}^2$ field at a $\Delta t = 10$ s cadence over ~50 continuous nights and $\sim 1.35 \times 10^6$ source stars with $g < 20$. Using a synthetic light-curve injection and recovery pipeline with finite-source treatment, we derived the survey detection efficiency and translated it into model-independent limits on the microlensing event rate, mass-dependent limits for a fast lens benchmark ($v_0 = 0.1c$), and projected constraints on the primordial black hole abundance toward the LMC. We also examined a structural limitation of the survey: because its bright magnitude limit ($g < 20$) selects intrinsically luminous, physically large sources, the monitored stars are systematically large on the sky and therefore maximally vulnerable to finite-source suppression in precisely the low-mass regime where the fast cadence would otherwise grant access. This effect sets the low-mass wall and explains why a shallow, bright-source-limited survey is more strongly affected by finite-source suppression than deeper surveys such as OGLE or Subaru-HSC at comparable lens masses. The same fast-cadence, wide-field staring strategy nonetheless provides a powerful complementary capability: by densely sampling every event in its field without external triggering, SkipperCam can improve the characterization of higher-order effects such as parallax and finite-source signatures, enabling precise per-event parameter measurements alongside the large statistical samples of wide-area surveys.

Beyond microlensing, the instrument's sub-electron noise and flexible readout modes make it a compelling platform for LSST-driven stellar occultation follow-up, where breaking the conventional trade-off between sub-second cadence and sensitivity is essential for resolving Fresnel-scale features of kilometer-scale outer Solar System bodies. Additional opportunities in fast photometric variability, transient detection, and blue-sensitive imaging further leverage the intrinsic strengths of the technology.

Finally, SkipperCam is designed as a risk-reduction pathfinder for a future degree-scale Skipper-CCD instrument. Its 20-CCD raft reproduces the mechanical and interface footprint of a 2×2 OTA module in the One Degree Imager, enabling direct integration within the WIYN 3.5 m focal plane. By validating the raft-scale focal-plane and electronics design, the MIDNA-based fast readout, the backside-processed blue quantum efficiency, and an early version of the ODI control and data-acquisition software in a realistic on-sky, survey-mode setting, the pathfinder retires the principal technical risks of the full ODI-Skipper program while delivering a self-contained scientific dataset. In doing so it represents a concrete first step toward a new observational regime defined by photon-limited precision, rapid cadence, and wide-field coverage, with broad implications for time-domain astrophysics, cosmology, and fundamental physics.

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A Detailed Survey Parameters

Parameter	Value
Telescope & instrument	
Telescope	Curtis Schmidt (CTIO)
Aperture	0.6 m
Readout noise	$7\ e^-$
Filter	g
Pixel scale	1.43 ”/pixel
Observing strategy	
Cadence, Δt	10 s
Night length	$\sim$ 8 hr
Number of nights	50
Target field & sources	
Target system	Large Magellanic Cloud
Field center (α, δ)	$(80.89°, -69.76°)$
Survey area, Ω	2.1 deg^2
Source distance, d_s	49.59 kpc
Monitored source stars, $N_\star$	1.35×10^6
Source catalogue	SMASH (DECam)
Source selection limit	$g < 20$
Red-clump reference, $I_{\rm RC}$	18.5 mag
Red-clump source size, $\theta_{*,\rm RC}$	0.91 μas

Table 1. Summary of the SkipperCam high-cadence LMC survey parameters.

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